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Title : JMeter Performance Testing & CI/CD Integration

1) Create the JMeter Test Plan : Create a JMeter test plan containing five Thread Groups.

2) Configure the Thread Groups

Configure each Thread Group with:

- Number of Users: **10**
- Ramp-Up Period: **5 seconds**
- Loop Count: **3**

3) Configure Test Scenarios: Add the required HTTP Request samplers, test data, parameters, listeners and assertions for each application also

4) Execute the Test in Non-GUI Mode: Execute the JMeter test using the command line to perform the load test efficiently.

5) Analyze Results Using Aggregate Graph/Report

Open the generated JTL file in JMeter and analyze the performance results using the Aggregate Graph/Report.

6) Validate Performance Metrics

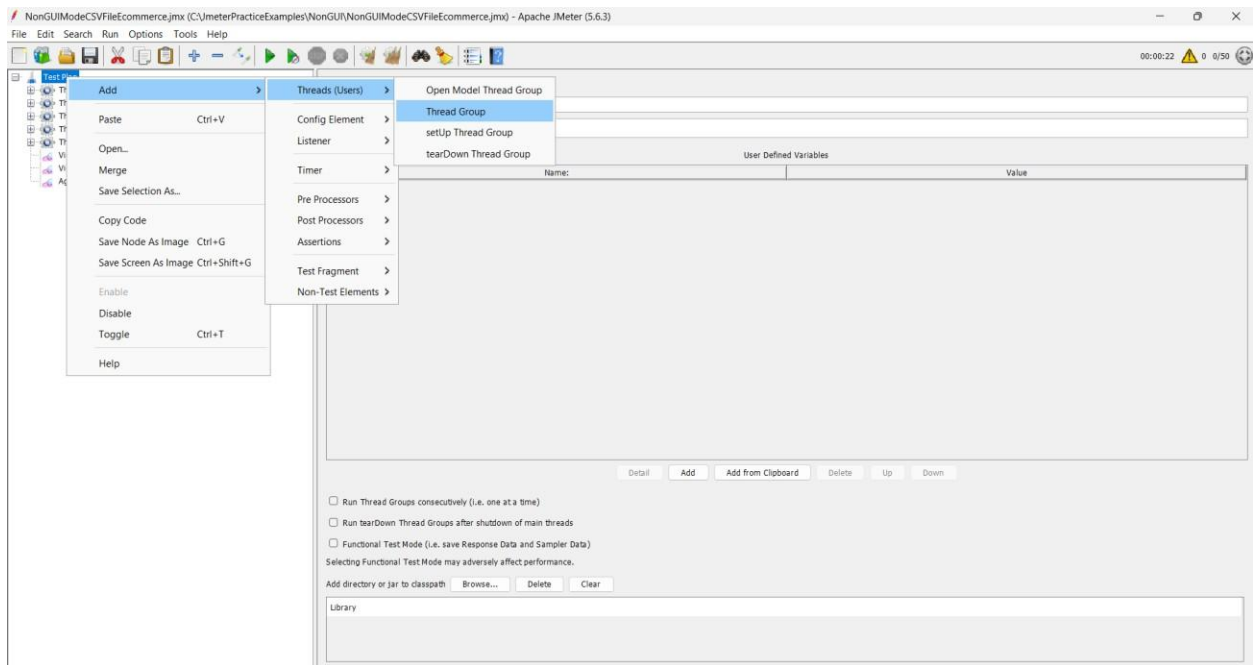
- Total Samples
- Throughput
- Latency
- Median
- Minimum Response Time
- Maximum Response Time
- 90th Percentile
- 95th Percentile
- Error Percentage

7) Generate the HTML Dashboard Report and Integrate with Jenkins

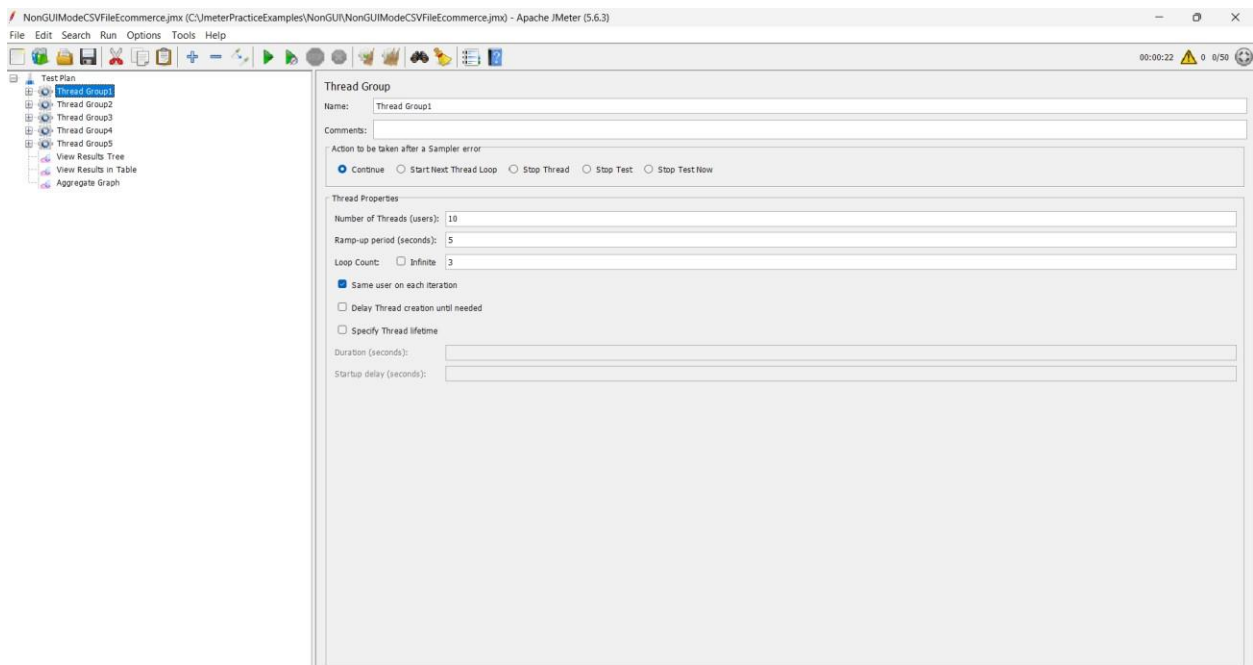
Use the JTL file to generate the JMeter **HTML Dashboard Report** containing graphical performance statistics.

8) Validate Jenkins Execution

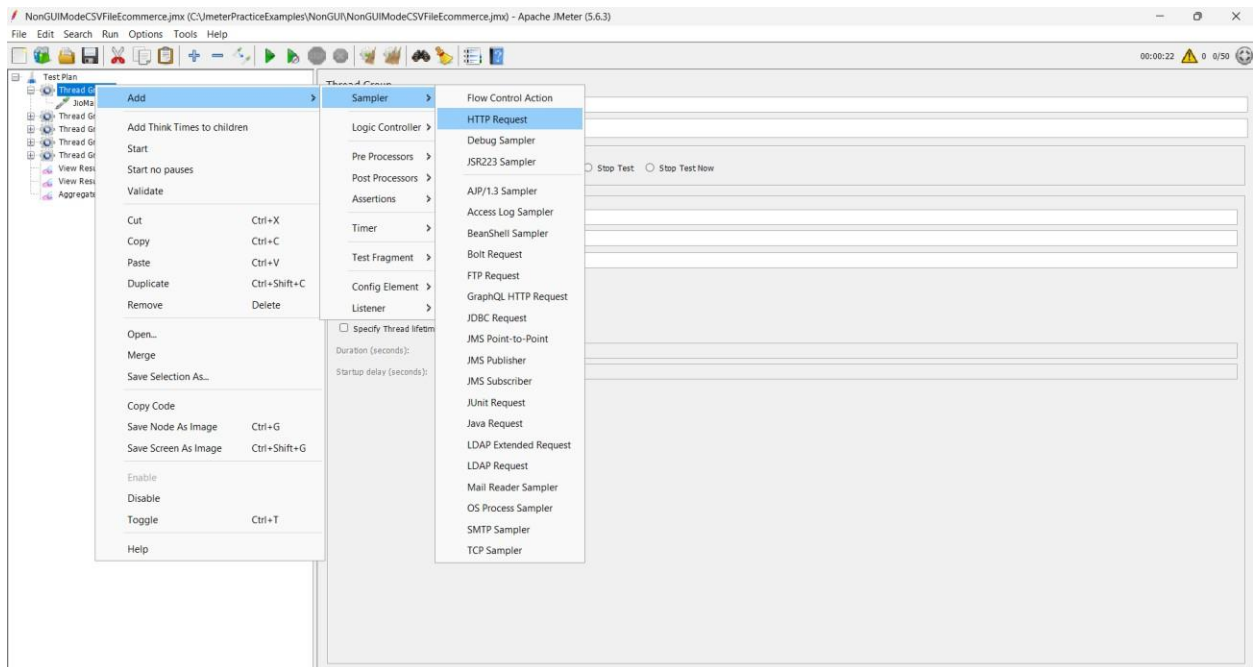
Step 1: Create a JMeter test plan containing five Thread Groups



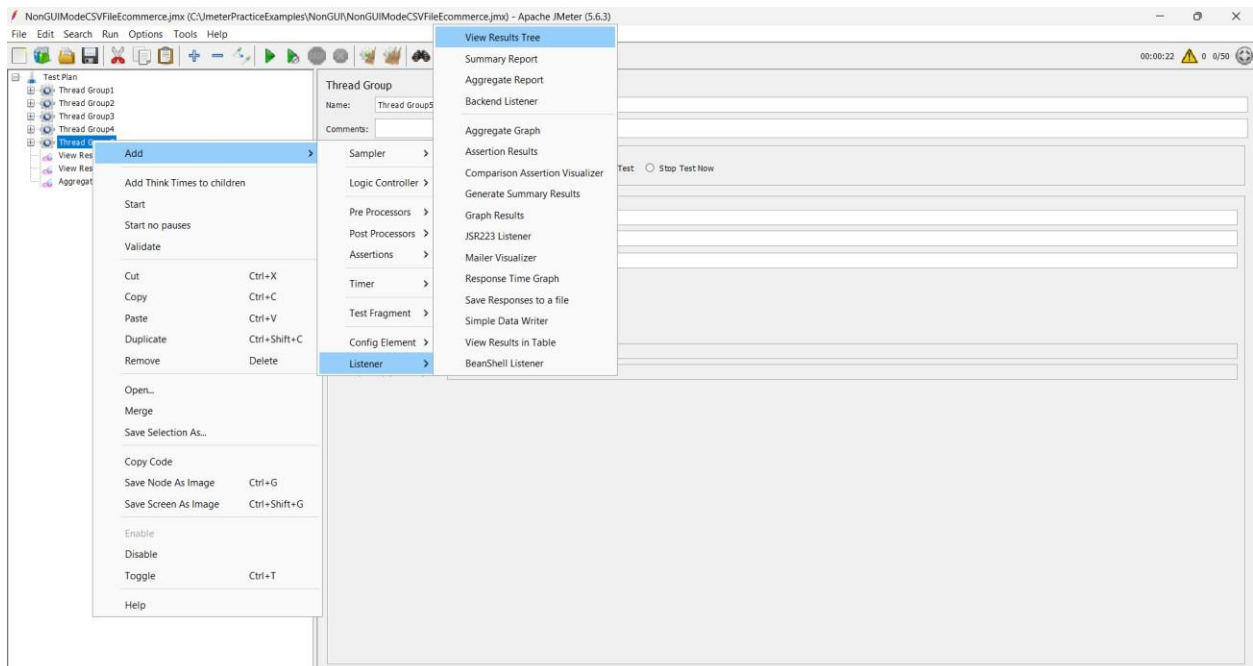
Step 2: Configure each Thread Group according to the requirements mentioned

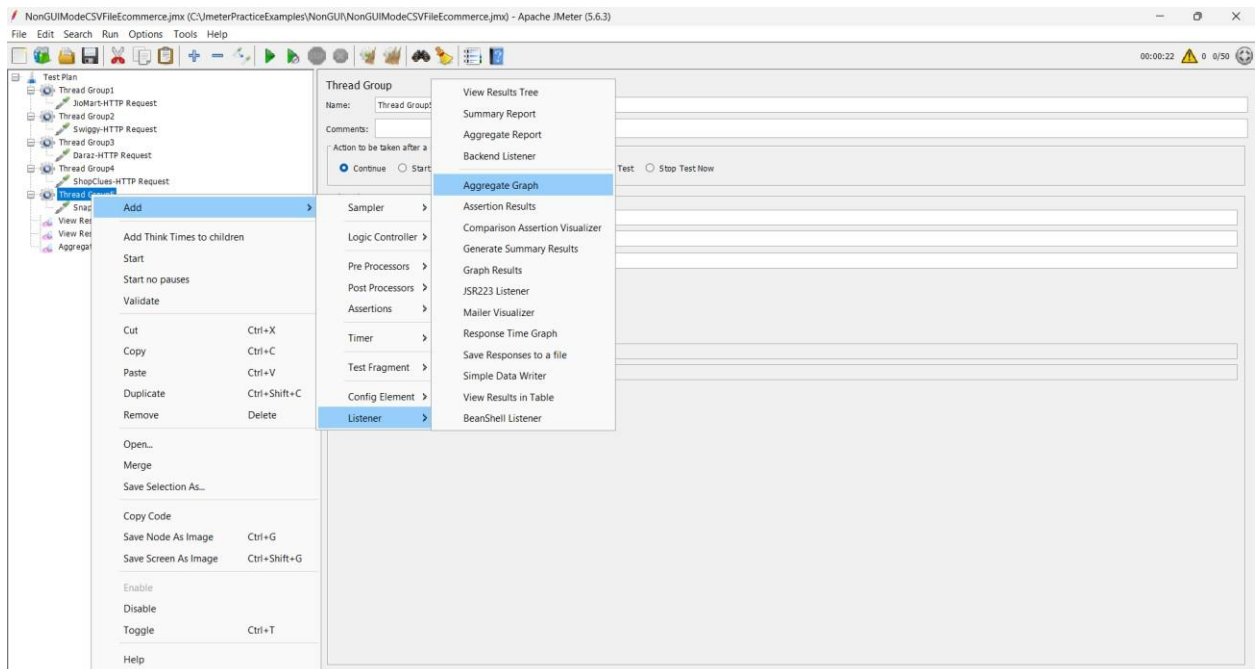
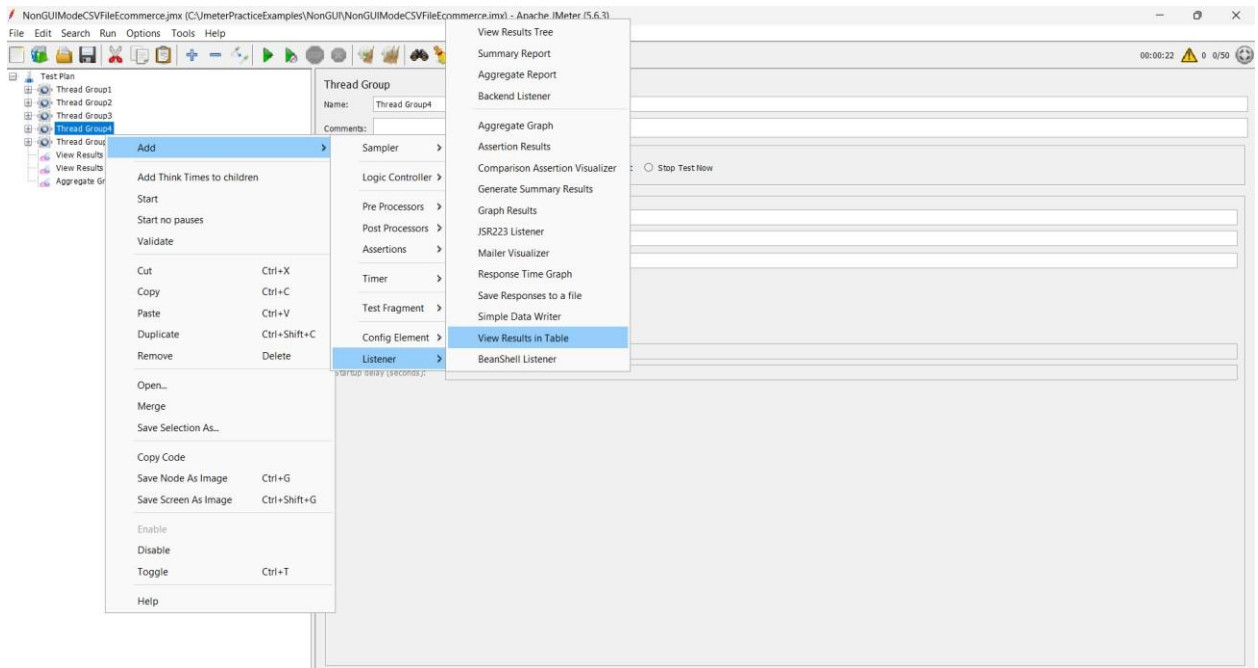


Step 3: Add samplers to each thread group using different E-commerce websites



Step 4: Add listeners





Step 5 : Run and validate

NonGUIModeCSVFileCommerce.jmx (C:\meter\PracticeExamples\NonGUI\NonGUIModeCSVFileCommerce.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan

- Thread Group1
 - JokMarr-HTTP Request
- Thread Group2
 - Swiggy-HTTP Request
- Thread Group3
 - Daraz-HTTP Request
- Thread Group4
 - ShopClues-HTTP Request
- Thread Group5
 - SnapDeal-HTTP Request
- View Results Tree
- Aggregate Graph

View Results in Table

Name: View Results in Table

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
1	18:19:03.410	Thread Group4 4-1	ShopClues-HTTP Req...	789	✓	316269	143	712	669
2	18:19:03.768	Thread Group4 4-2	ShopClues-HTTP Req...	436	✓	316270	143	254	312
3	18:19:04.201	Thread Group4 4-1	ShopClues-HTTP Req...	48	✓	316268	143	19	0
4	18:19:04.250	Thread Group4 4-1	ShopClues-HTTP Req...	40	✓	316268	143	20	0
5	18:19:04.205	Thread Group4 4-2	ShopClues-HTTP Req...	110	✓	316267	143	34	0
6	18:19:04.316	Thread Group4 4-2	ShopClues-HTTP Req...	79	✓	316266	143	25	0
7	18:19:03.410	Thread Group5 5-1	SnapDeal-HTTP Requ...	1063	✓	386397	154	1016	647
8	18:19:04.268	Thread Group4 4-3	ShopClues-HTTP Req...	211	✓	316267	143	136	94
9	18:19:04.474	Thread Group5 5-1	SnapDeal-HTTP Requ...	106	✓	386381	154	32	0
10	18:19:04.480	Thread Group4 4-3	ShopClues-HTTP Req...	124	✓	316267	143	54	0
11	18:19:04.580	Thread Group5 5-1	SnapDeal-HTTP Requ...	111	✓	385921	154	41	0
12	18:19:04.605	Thread Group4 4-3	ShopClues-HTTP Req...	119	✓	316267	143	47	0
13	18:19:03.767	Thread Group5 5-2	SnapDeal-HTTP Requ...	1218	✓	386078	154	823	290
14	18:19:04.269	Thread Group5 5-3	SnapDeal-HTTP Requ...	734	✓	386115	154	515	55
15	18:19:03.764	Thread Group3 3-2	Daraz-HTTP Request	1335	✓	228574	311	443	346
16	18:19:04.766	Thread Group5 5-4	SnapDeal-HTTP Requ...	390	✓	385959	154	162	111
17	18:19:03.410	Thread Group3 3-1	Daraz-HTTP Request	1772	✓	228605	311	796	700
18	18:19:05.003	Thread Group5 5-3	SnapDeal-HTTP Requ...	211	✓	385859	154	69	0
19	18:19:05.156	Thread Group5 5-4	SnapDeal-HTTP Requ...	216	✓	385867	154	82	0
20	18:19:04.765	Thread Group4 4-4	ShopClues-HTTP Req...	635	✓	316267	143	339	196
21	18:19:05.214	Thread Group5 5-3	SnapDeal-HTTP Requ...	234	✓	385867	154	97	0
22	18:19:05.372	Thread Group5 5-4	SnapDeal-HTTP Requ...	244	✓	385867	154	101	0
23	18:19:04.985	Thread Group5 5-2	SnapDeal-HTTP Requ...	713	✓	385859	154	75	0
24	18:19:05.400	Thread Group4 4-4	ShopClues-HTTP Req...	351	✓	316267	143	134	0
25	18:19:04.266	Thread Group3 3-3	Daraz-HTTP Request	1640	✓	228610	311	302	171
26	18:19:05.751	Thread Group4 4-4	ShopClues-HTTP Req...	271	✓	316268	143	113	0
27	18:19:05.268	Thread Group5 5-5	SnapDeal-HTTP Requ...	780	✓	385867	154	348	213
28	18:19:03.761	Thread Group2 2-2	Swiggy-HTTP Request	2395	✓	355213	126	1921	296
29	18:19:06.048	Thread Group5 5-5	SnapDeal-HTTP Requ...	329	✓	385867	154	145	0
30	18:19:05.267	Thread Group4 4-5	ShopClues-HTTP Req...	1234	✓	316268	143	581	365
31	18:19:05.099	Thread Group3 3-2	Daraz-HTTP Request	1411	✓	228590	311	169	0
32	18:19:05.182	Thread Group3 3-1	Daraz-HTTP Request	1470	✓	228603	311	183	0
33	18:19:06.378	Thread Group5 5-5	SnapDeal-HTTP Requ...	286	✓	385867	154	185	0

☐ Scroll automatically? ☐ Child samples? No of Samples 150 Latest Sample 1083 Average 2924 Deviation 3018

NonGUIModeCSVFileCommerce.jmx (C:\meter\PracticeExamples\NonGUI\NonGUIModeCSVFileCommerce.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

00:00:22 0 0/50

Test Plan

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- Thread Group5
 - SnapDeal-HTTP Request
- View Results Tree
- Aggregate Graph

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

Search: Case sensitive Regular exp. Search Reset

Text

Sampler result

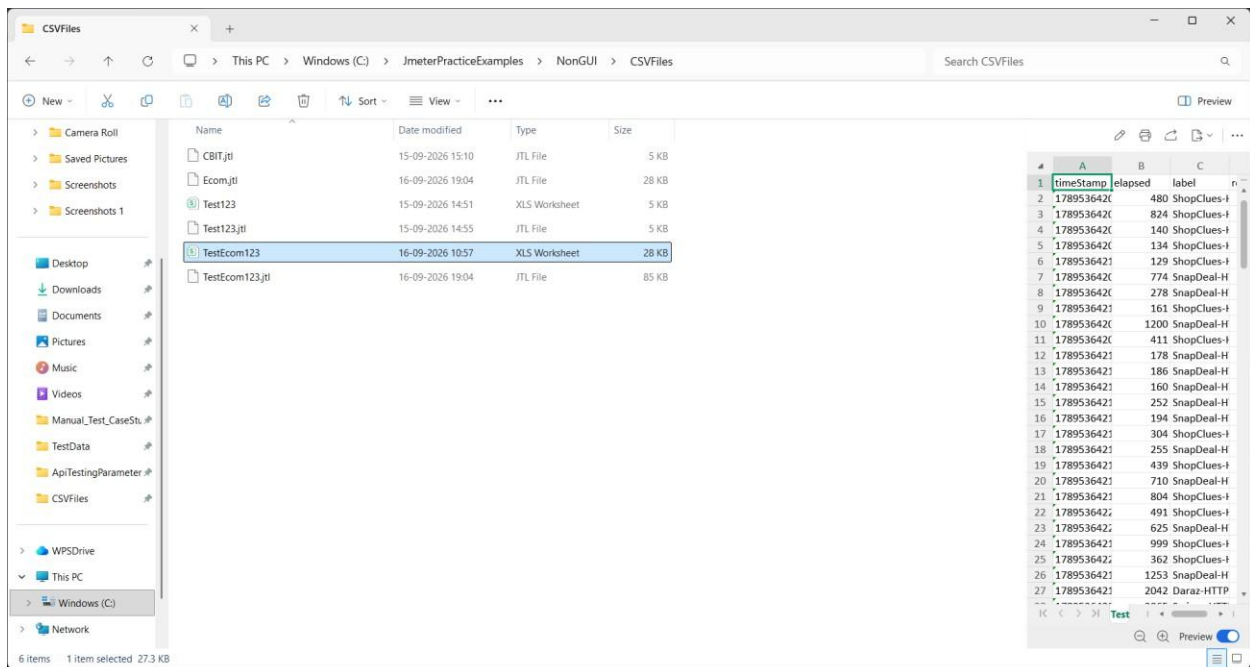
- ShopClues-HTTP Request
- ShopClues-HTTP Request
- ShopClues-HTTP Request
- ShopClues-HTTP Request
- ShopClues-HTTP Request
- SnapDeal-HTTP Request
- ShopClues-HTTP Request
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- Daraz-HTTP Request
- SnapDeal-HTTP Request
- Swiggy-HTTP Request
- SnapDeal-HTTP Request
- ShopClues-HTTP Request
- Daraz-HTTP Request
- SnapDeal-HTTP Request

☐ Scroll automatically? Raw Parsed

Step 6:

1. Open Command Prompt.
2. Navigate to the JMeter bin folder using the following command:
3. `cd /d "C:\ApacheJmeter5.6.3\apache-jmeter-5.6.3\bin"`
4. Generate the CSV result file by executing the JMeter test plan in non-GUI mode:
`jmeter -n -t`
"C:\JmeterPracticeExamples\NonGUI\NonGUIModeCSVFileEcommerce.jmx" -l
"C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.csv"

The test execution results are saved in the specified location as a CSV file.

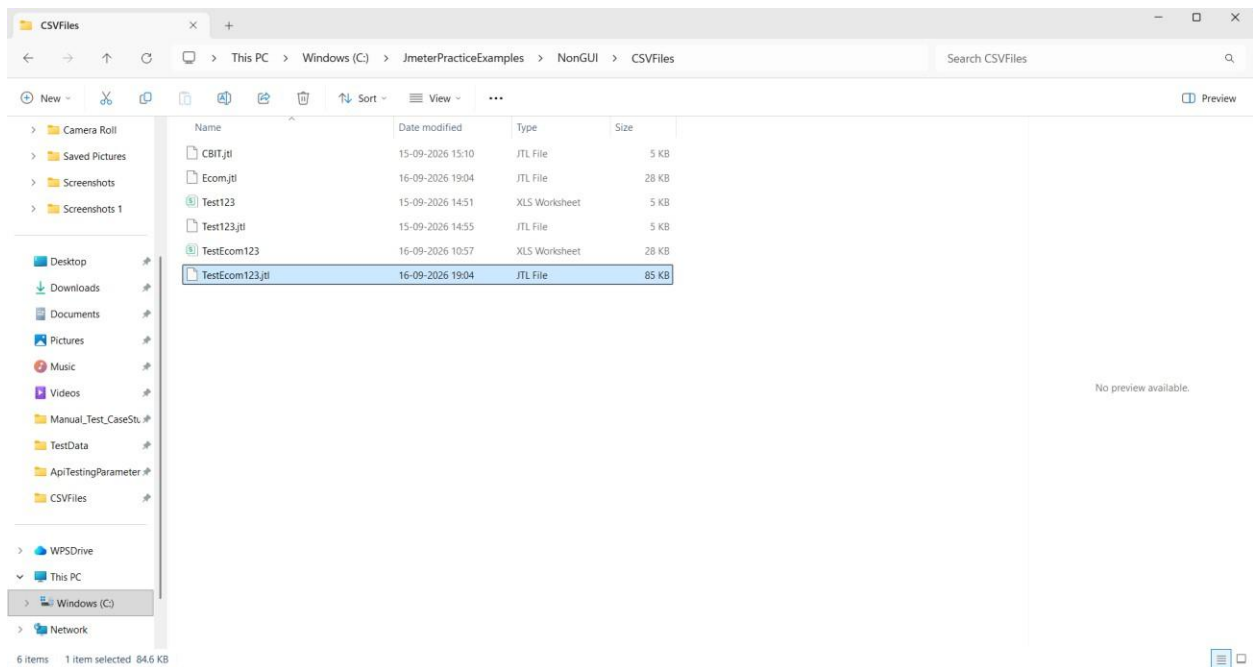


The screenshot shows a Windows File Explorer window titled 'CSVFiles' with the address bar showing the path 'This PC > Windows (C:) > JmeterPracticeExamples > NonGUI > CSVFiles'. The file list contains several files, with 'TestEcom123.csv' selected. A preview window on the right displays the CSV data.

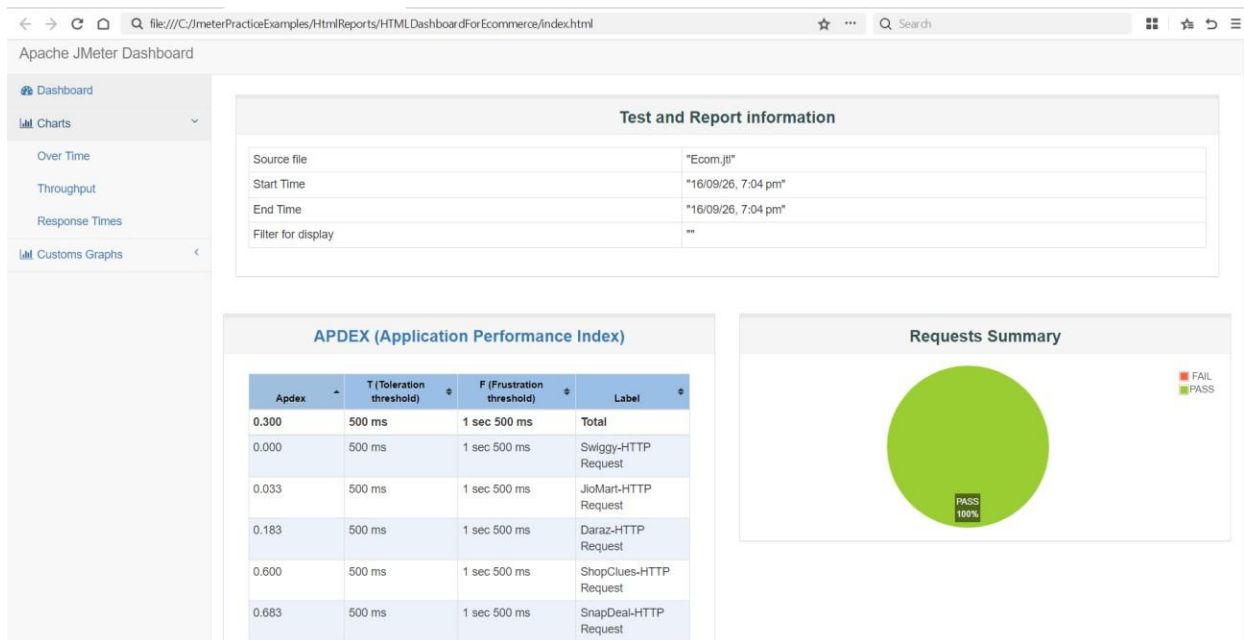
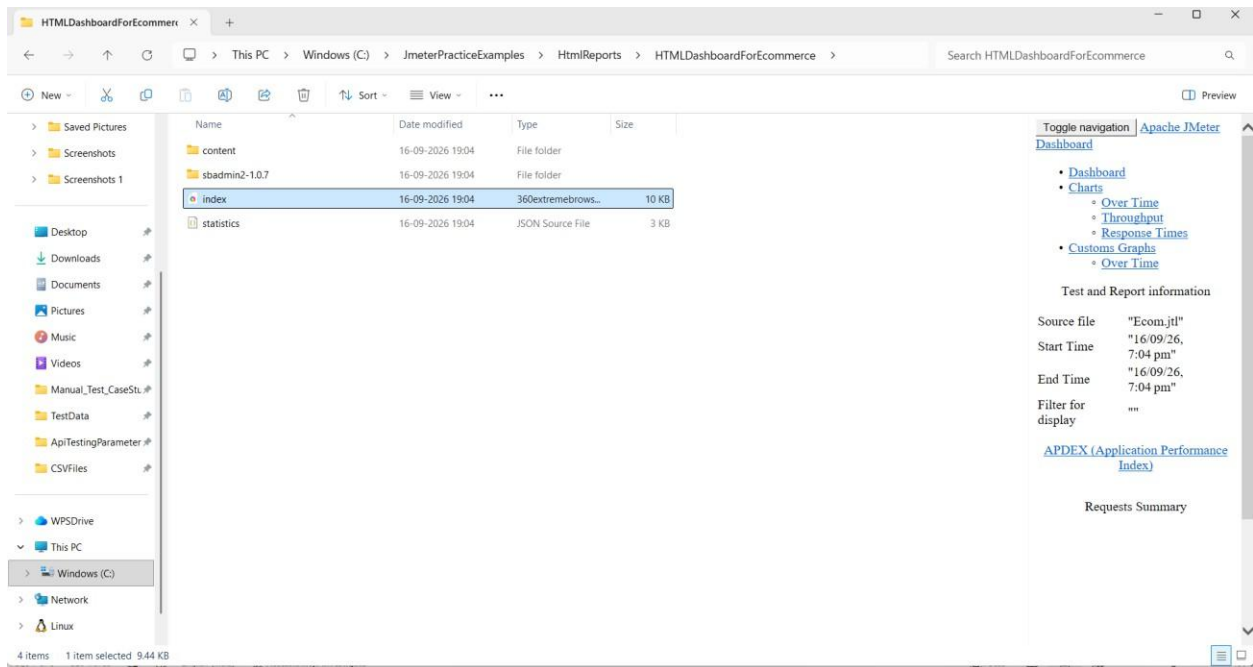
#	A	B	C
1	timestamp	elapsed	label
2	1789536420	480	ShopClues-H
3	1789536420	824	ShopClues-H
4	1789536420	140	ShopClues-H
5	1789536420	134	ShopClues-H
6	1789536420	129	ShopClues-H
7	1789536420	774	SnapDeal-H
8	1789536420	278	SnapDeal-H
9	1789536420	161	ShopClues-H
10	1789536420	1200	SnapDeal-H
11	1789536420	411	ShopClues-H
12	1789536420	178	SnapDeal-H
13	1789536420	186	SnapDeal-H
14	1789536420	160	SnapDeal-H
15	1789536420	252	SnapDeal-H
16	1789536420	194	SnapDeal-H
17	1789536420	304	ShopClues-H
18	1789536420	255	SnapDeal-H
19	1789536420	439	ShopClues-H
20	1789536420	710	SnapDeal-H
21	1789536420	804	ShopClues-H
22	1789536420	491	ShopClues-H
23	1789536420	625	SnapDeal-H
24	1789536420	999	ShopClues-H
25	1789536420	362	ShopClues-H
26	1789536420	1253	SnapDeal-H
27	1789536420	2042	Daraz-HTTP

Step 7:

1. Generate the JTL file to use as the input for the Aggregate Graph: `jmeter -n -t "C:\JmeterPracticeExamples\NonGUI\NonGUIModeCSVFileEcommerce.jmx" -l "C:\JmeterPracticeExamples\NonGUI\CSVFiles\TestEcom123.jtl"`
2. Open the generated JTL file in JMeter and use the Aggregate Graph/Report to analyze the performance results, including Throughput, Latency, Median, Minimum, Maximum, Percentiles, Total Samples, and Error Percentage.



Step 7: Html reports



Step 9 : Configure Jenkins

Jenkins / JenkinsCICDUsingJmeter / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command ?

Command

See the list of available environment variables

```
cd C:\ApacheJmeter5.6.3\apache-jmeter-5.6.3\bin
jmeter -njmeter.save.saveservice.output_format=xml -n -t
"C:\JmeterPracticeExamples\CICDPracticeTests\NonGUIModeCSVFileEcommerce.jmx"
-l "C:\JmeterPracticeExamples\CICDPracticeTests\JenkinsCICD.jtl"
```

Advanced

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Publish Performance test result report ?

Save Apply

Step 10: Analyze the performance Trend

Jenkins / JenkinsCICDUsingJmeter / Performance Trend

Performance Trend

Last Report
Filter trend data

Test file: JenkinsCICD.jtl

Response time

ms

60,000
50,000
40,000
30,000
20,000
10,000
0

90% line 95% line average median

Percentage of errors

%

100
80
60
40
20
0

errors

Trend report

Builds

Filter

Today

- #3 1:54 pm
- #2 11:47 am
- #1 11:44 am

Jenkins 2.568.2



Jenkins

JenkinsCICDUsingJmeter

Performance Trend

Trend report



Trend report: JenkinsCICD.jtl

Build	Line 90.0(ms)	Line 95.0(ms)	Max(ms)	Median(ms)	Min(ms)	Percentage of errors	average	average kb	errors	total kb
#2	26267	65067	179643	4495	357	1	11833	1606.88	1	241031.39
#3	19997	42468	179643	4923	228	0	10139	1608.68	1	482603.23

Jenkins 2.568.2

Step 11: JMeter Aggregate Graph – Performance Test Results

